

Confidentiality Notice

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File scanner

Your task is to design and implement a file scanner. The scanner tags files as sensitive, if their content contains a sensitive word, provided as an input.

Input

1. Full path to a file/folder in your local file system (string)
2. Sensitive word (string)

Functional requirements

1. The scanner should support the following file extensions (other files can be skipped)
 - a. .txt
 - b. .json
 - c. .csv
 - d. .docx
 - e. .pdf
2. If the file contains the sensitive word, print a proper message with the full file path.
3. If the given path leads to a folder, the scanner should scan all the files and folders within that folder, recursively.
4. The scanner should support very large files that cannot be read into memory at once.
5. The scanner should scan vast and deep directory trees efficiently.
6. The scanner supports case insensitive, exact word matching. For example, if the sensitive word is **goat** :

File name	Text	Verdict	Comments
a.txt	Does a goat eat meat?	MATCH	
b.txt	GOAT is an abbreviation for Greatest Of All Time: used to refer to or describe the person who has performed better than anyone else ever, especially in a sport.	MATCH	Case insensitive match
c.txt	Riddle: You go at red but stop at green. What am I? Answer: Watermelon!	NO MATCH	go at != goat
d.txt	The main part of a goat's diet is called roughage.	NO MATCH	goat's != goat

Assumptions

1. The sensitive word is relatively short, and can be stored as-is in memory.
2. Single line can be stored as-is in memory
3. There's no need to find more than one match in a given file.
4. If the path doesn't exist, print a proper error message and quit the program.

Expectations

1. After you evaluate all the requirements, you should define an MVP (Minimum Viable Product) according to the time you have and your effort estimation.
2. You should implement a fully working solution according to the MVP definition, in any high-level programming language.
3. Your code should be clear and extensible.

Good luck!

